



Software Engineering

CSC443-A | Fall, 2025 | 4 Credits

Augsburg University

[NOTE: All course policies are subject to change with notice. - Last updated August 20, 2025]

COURSE INFORMATION

Class Meeting Day and Times: **Tuesdays and Thursdays 8:30 - 10:00 AM**

Class Location: **Hagfors Center 367 - Computer Lab**

Prerequisites, if Any:

- **CSC321 - Software Design and Development**
- **CSC341 - Data Structures and Algorithms**
- **Coding-intensive experience - 1 of**
 - **CSC311 - Web Applications and Databases**
 - **CSC351 - Algorithms**
 - **CSC399 - Internship**
 - **CSC421 - Mobile Computing**
 - **CSC431 - Introduction to AI Robotics**
 - **DST314 - Programming for Data Science**
 - **MIS476 - Information Systems Projects**

IMPORTANT DATES

- Last day to drop without a 'W' grade & 100% refund - **Wednesday, September 10, 2025**
- Last day to withdraw from all fall classes and receive a 50% refund - **Friday, October 17, 2025**
- Last day to withdraw - Full Semester class - **Friday, November 7, 2025**
- Last day to designate grading option (undergraduate courses) - **Tuesday, December 30, 2025**

INSTRUCTOR INFORMATION

Instructor: **Dr. Nimanthi Atukorala**

Office: Hagfors Center 369F

Office Hours: Mondays & Fridays (in person) 10:50 am - 11:50 am.

Thursdays (on Zoom) 1:00 pm - 2:00 pm.

E-mail: atukoral@augsborg.edu

Preferred Pronouns: She/Her/Hers

COURSE OVERVIEW & OBJECTIVES

The goal of this course is to provide an experience with good Software Engineering practices that allow creating and maintaining intuitive, useful, well-tested, and accessible software systems. The course is designed to introduce major Software Engineering concepts in different stages of software development. You will learn about and apply common design patterns and good development practices for team-based large-scale projects. Communication is key for team projects, thus you will practice various forms to describe and document software design, such as UML diagrams, use cases, and code documentation. We will discuss professional responsibility and the impact of software and computer technology on society.

The semester-long project for the CSC 443 students will provide a real-world software development experience in a team environment.

The main objectives of this course are,

- Improve students' problem-solving ability. This includes being able to identify the key aspects of a problem, break the problem down into smaller units (in many different ways), and develop an understanding of how to assemble a larger system from the output of smaller (sub)systems.
- Describe the characteristics of quality software, some examples include correct, extendable, maintainable, robust, secure, testable, and efficient.
- Identify common software design problems and the applicable design pattern(s), their strengths, and weaknesses.
- Use best practices and professional tools for design, development, testing, and documentation. Some example tools include version control, debuggers, testing frameworks, linters, UML, and auto-generators for documentation.
- Gain experience in planning and executing a large project in a small group.
- Communicate verbally and in writing to effectively describe and justify software solutions for either a technical or non-technical audience.
- Understand professional responsibility in software development, including its social impact and ethical issues along various dimensions, such as economic disruption, bias, job displacement, mental health, equitable access, and privacy.

As senior keystone course:

- Reflect on the concept of vocation and how Software Engineering can apply/connect to vocation.
- Think critically about how what problems you choose to solve (and the way you choose to solve those problems) affect society and reflect how the Augsburg Mission ties into all this.
- Reflect on your entire Augsburg Education and how your Augsburg experience has affected the type of problems you want to solve and how you solve those problems.

TEXTBOOK(S) & COURSE MATERIALS

CSC 443 Recommended Textbook:

Software Engineering by Ian Sommerville

ISBN: 9780133943030

Publisher: Pearson Education, Inc.

Edition: 10th Edition, 2016

Software [free open-source]

- Access to a Java development environment.

Technology requirements for the class

- Check email, access Moodle resources including downloading PDFs, and post PDFs to Moodle.
- Attend live remote classes on Zoom, ideally with audio/video on, and take proctored quizzes over Zoom with video if possible.
- Have multiple documents and multiple applications open simultaneously during class: Zoom, Google Jamboards, PDFs, etc.
- Contribute to a collaborative virtual whiteboard (Google Jamboard) by writing on an electronic surface. Alternatively, complete collaborative work by typing on virtual sticky notes and/or talking while a team member scribes for you.

To complete these tasks,

- You need a computer or tablet with a camera and a microphone for the class. If you do not own one, we encourage you to borrow a computer from the university for the semester.
- You need reliable Internet access for class. We encourage you to find a place to attend a class where you will feel comfortable sharing your video and audio at least some of the time. We understand if that is not possible.
- Please contact your instructor or your faculty mentor/academic advisor if you need help acquiring or using these resources or if you need help finding a place to work where you can comfortably share the video as needed.

COURSE POLICIES

Attendance and Participation

Faithful attendance is an absolute requirement for this course. Excessive absences will seriously distract you from both your success and the class as a whole. In addition to being present, you need to engage with the course content and fellow students.

Many of the class periods will include time for project and or assignment work/refactoring and you need to be present and contributing!

If you will be absent from class, please notify me as soon as possible (preferably by email so we both have a record of exactly when you contacted me). If something is going on where you need to miss class, please come talk with me.

If for any reason you are unable to attend class regularly and/or are experiencing difficulty in either keeping up with or understanding the course workload or course structure, please speak with me immediately.

It is your responsibility to come to class and know your obligations for all assignments. If you do miss class, it is your responsibility to get notes/tasks from one of your classmates.

Important Note regarding Attendance: Federal law requires faculty to report students who never show up or who stop attending class. This report may cause a recalculation in the student's financial aid award, possibly requiring the student to return funds already received.

Academic Integrity

It is assumed that all students at Augsburg University will conduct themselves ethically and professionally. The student guide, including the conduct code, should guide your behavior.

Academic honesty (Academic Honesty Policies) is integral to your success in this course and at Augsburg University. The following is an excerpt from the policy:

A University is a community of learners whose relationship relies on trust. Honesty is necessary for the functioning of the Augsburg University community and dishonesty is, therefore, abhorred and prohibited.

One example of how trust is destroyed by a particular form of dishonesty is found in plagiarism and its effects. In its 1990 'Statement of Plagiarism,' the American Association of University Professors (AAUP) Committee B on Professional Ethics notes that one form of academic dishonesty, plagiarism, 'is theft of a special kind [in which] a fraud is committed upon the audience that believes those ideas and words originated with the deceiver. Plagiarism is not limited to the academic community but has, perhaps, its most pernicious effect in that setting. It is the antithesis of the honest labor that characterizes true scholarship and without which mutual trust and respect among scholars is impossible.'

...Penalties imposed by the professor may include a 'zero' or failing grade on the assignment or examination which involved dishonesty, other academic penalties as outlined in the syllabus for the course or other statement of policies distributed by the professor, forced withdrawal from the course, or failure in the course.

When you use someone else's work you must be sure you are not committing plagiarism or violating copyright. If you need assistance with proper citations, you may visit the Writing Lab, The Gage Center for Student Success, or the Lindell Library. I prefer MLA or Chicago Style, however, I will accept APA.

Students who are found in violation of the academic honesty policies about plagiarism and copyright infringement will be reported. I reserve the right to lower your grade if you violate the Academic Honesty policy in this course. The student may also face further penalties as determined by the appropriate Dean or university administrator.

Disability Accommodations and Accessibility

Augsburg University is committed to providing an environment where all students have the opportunity to equally participate in the academic experience, including students with disabilities. Students with disabilities have rights as determined by federal and state laws which require institutions to provide reasonable accommodations for the student's disability to afford an equal opportunity to participate in the university's programs, courses, and activities.

If you need an accommodation because of a disability, please contact The Center for Learning and Adaptive Student Services (CLASS) at 612-330-1053, class@augsborg.edu, or stop by the Gage Center welcome desk on the link level of the Lindell Library to discuss how accommodations may be implemented in your Augsburg courses. If you have a letter from this office indicating you require academic accommodations, please be sure to check in with your instructor to ensure that you get the accommodations you need in this class.

Statement of Inclusion and Participation

Augsburg University values the diversity of persons, perspectives, and convictions. Critical thinking, rigorous analysis, and open discussion across a full range of ideas lie at the heart of the University's mission as an institution of higher learning. Essential to the University in living out its mission of educating students to be "informed citizens, thoughtful stewards, critical thinkers, and responsible leaders" is that the foundation be one of diversity, inclusion, equity, and intercultural competency.

We acknowledge that racism, classism, ableism, homophobia, transphobia, sexism, and other forms of oppression exist. In this class, people of all races, ethnicities, gender expressions and gender identities, religions, ages, sexual orientations, abilities, socioeconomic backgrounds, political affiliations, regions, and nationalities are strongly encouraged to share their rich array of perspectives and experiences. In doing so, we will be asked to take responsibility for our privileges and examine our words and actions. When we know better, we can do better.

Sexual Misconduct, Required Reporting, and Title IX

Augsburg University is committed to combating sexual misconduct. As a result, you should know that University faculty and staff members are required to report any instances of sexual misconduct, including harassment and sexual violence, to the University's Title IX Coordinator so that the student may be provided with appropriate resources, support options, and reporting options. What this means is that, as your professor, I am required to report any incidents of sexual misconduct that are directly reported to me, or of which I am somehow made aware.

There are two important exceptions to this requirement:

1. There are confidential resources in the Center for Wellness and Counseling (CWC) and Campus Ministry that do not have this reporting responsibility and can maintain confidentiality. More information can be found on page 15 of the Sexual Misconduct Policy: <http://inside.augsburg.edu/studentaffairs/harassment-and-sexual-harassment-reporting-form/>
2. An important exception to the reporting requirement exists for academic work. Disclosures about sexual misconduct that are shared as part of an academic project, classroom discussion, or course assignment, are not required to be disclosed to the University's Title IX office.

If you have questions or concerns, I encourage you to contact Ann Garvey, Title IX Coordinator, 612-330-1168, or garvey@augsborg.edu

COURSE WORKLOAD

Plan on spending 10-12 hours per week outside of class time working on the course.

In-class Activities:

The in-class activities include individual or group work during class time, mainly focused on the software engineering concepts learned that day.

Written Homework:

Throughout the semester, you will be assigned written homework related to software design, development, and professional responsibility in software development. Written homework is typically independent work.

Exams:

There will be two written exams (mid-term and final). Written exams are designed to give you the opportunity to show me that you have acquired and sufficiently understood the terminology and concepts covered in this course.

A typical written exam will consist of 5 – 10 questions designed to probe your understanding of the content and how to apply your knowledge. Questions may include (but are NOT limited to): short answers, essays, reading/refactoring/producing diagrams, breaking down a problem, critiquing a software project, and assessing competing solutions.

Semester-long Team Project:

Over the semester, you will be working on a team project. The purpose of the projects is for you to gain real experience DOING Software Engineering using many of the concepts we will encounter during the course.

You will need to spend a significant amount of time outside of class working on your projects to complete them. There will be class time allotted for working on your project primarily for management-type activities and for you to demonstrate your progress and participation to me.

Projects will require you to showcase your projects to your classmates as well as receive and provide peer review.

The project will follow the Agile development approach. A typical sprint duration will be 2 weeks. The team leader role will be rotated among team members. Each team member is required to hold the team leadership for at least one sprint.

- **Project Participation:** Each member is required to actively participate and collaborate in the team project.
- **Project Deliverables:**
 1. Project Proposal
 2. Team initiation
 3. Sprint reports: At the end of each sprint you are required to submit a document summarizing the current progress.

A typical sprint report includes two parts:

Group component: The current team leader must submit a report on the responsibilities assigned to each team member for that particular sprint, a summary of team meetings (SCRUM meetings), and a summary of updates to the SCRUM board. This report must be shared among all the team members.

Individual component: Each team member must submit an individual progress report that includes (but is not limited to) your contribution to the team project during the sprint, your timeline of completing tasks, the challenges faced, and your goal for the next sprint. This report may require to include additional design plans such as use-case diagrams and class diagrams.

4. Sprint progress presentations: At the end of each sprint, each team is required to make a short presentation about the progress of the project during that sprint. The sprint leader must lead the presentation and each team member must participate and explain their contribution.
5. Midterm project-progress presentation
6. Final project demonstration
7. Final project user manual and final project reflection
8. Final project code

During this semester-long team project, it is more likely that you will have to self-learn new technologies, handle software installation problems, and work independently.

- **The instructor reserves the right to lower your final course grade if your project participation or collaboration falls well below expectations. Regardless of accumulated points, receiving less than 70% for the Project category will automatically result in a D grade or lower.**

LATE WORK POLICY [Subject to change with notice]

In-class activities are due by the end of class (unless otherwise noted) and they will not be accepted late.

Due time and date for projects and homework will be posted on Moodle.

A project deliverable or homework will be accepted until three days after the original due date with a 30% penalty and four to seven days with a 50% penalty. Any submission after seven days of the deadline will receive an automatic zero grade.

**No more than ONE late project deliverables are allowed during the semester.
No more than 2 homework assignments are allowed during the semester.**

EVALUATION & GRADING

You will be graded on the successful completion of all the in-class activities, homework, semester-project, exams, and participation.

DESCRIPTION	PERCENTAGE
Semester-long project work	50%
Class participation and in-class activities	20%
Written homework	20%
Exams (Midterm, Final)	10%
Total	100%

I use rubrics for major assignments and the Augsburg grade scale as the tentative grade scale.

Grades will be posted in the “Grades” area of our Moodle course. Typically, grades will be posted **one week** following the completion of an activity.

If you have questions about the grading, please contact me.

AUGSBURG UNIVERSITY GRADE SCALE			
Grade	Grade Points	Percentage	Description
A	4.00	93-100	Excellent
A-	3.67	90-92.99	
B+	3.33	87-89.99	
B	3.00	83-86.99	Undergraduate: Good Graduate: Satisfactory
B-	2.67	80-82.99	
C+	2.33	77-79.99	
C	2.00	73-76.99	Undergraduate only: Satisfactory

C-	1.67	70-72.99	
D+	1.33	67-69.99	
D	1.00	63-66.99	Poor
D-	0.67	60-62.99	
F	0.00	0-59.99	Failure
P	0.00		Pass*
N	0.00		No Pass
*The letter grade equivalent for a "P" grade is a C-/70 or higher for undergraduate courses and a B/83 or higher for graduate courses.			

- **Regardless of accumulated points, receiving less than 70% for the Project category will automatically result in a D grade or lower.**
- Please note that a C- or higher is required for a pass under the pass / no-pass grading option. A passing grade, C- or higher, is required for the course to apply to the Computer Science major or minor or the Mathematics major.

Disclaimer: This syllabus is intended to suggest the outline of the course; it is not absolute. Changes to the syllabus will be announced in class and/or displayed on Moodle.

Tentative Weekly Schedule

<i>Week</i>	<i>Lesson Overview</i>	<i>Homework/Project</i>
1 (Sep 03 - 05)	• <i>Introduction to the Course and Expectations</i>	<i>Written homework 01 out</i> <i>Project teams</i>
2 (Sep 08 - 12)	• <i>Software Development Life Cycle (SDLC) Models</i>	<i>Written homework 01 due</i> <i>Project proposals start</i>
3 (Sep 15 - 19)	• <i>Software Development Life Cycle (SDLC) Models (continue)</i>	<i>Written homework 02 out</i>
4 (Sep 22 - 26)	• <i>Software Requirements Engineering</i>	<i>Project proposals Due</i>
5 (Sep 29 - Oct 03)	• <i>Software Designing and Modeling</i>	<i>Sprint 1 starts</i> <i>Written homework 02 due</i>
6 (Oct 06 - 10)	• <i>Architectural Considerations</i>	<i>Written homework 03 out</i>

7 (Oct 13 - 17)	• <i>Design and Implementation</i> • <i>Augsburg Career Fair: Wed, Oct 15 from 11am - 1pm (Christensen Center Lobby)</i>	<i>Sprint 1 due</i>
8 (Oct 20 - 24)	• <i>Design and Implementation (continue)</i> • <i>Mid-Term Exam</i>	<i>Written homework 03 due</i> <i>Sprint 2 starts</i> <i>Midterm - Exam</i>
9 (Oct 27 - 31)	• <i>Midterm Project Presentations</i>	
10 (Nov 03 - 07)	• <i>Software Testing and Quality Assurance</i>	<i>Written homework 04 starts</i> <i>Sprint 2 due</i>
11 (Nov 10 - 14)	• <i>Software Maintenance and Refactoring</i>	<i>Sprint 3 starts</i>
12 (Nov 17 - 21)	• <i>Software Project Management</i>	<i>Written homework 04 due</i>
13 (Nov 24 - 28)	• <i>Trends in Software Engineering</i>	<i>Sprint 3 due</i>
14 (Dec 01 - 05)	• <i>Trends in Software Engineering (continue)</i>	<i>Sprint 4 starts</i>

<i>15</i> <i>(Dec 08 - 12)</i>	• <i>Course Reflection</i> • <i>Final Project Presentations</i>	<i>Sprint 4 due</i>
<i>Final Exam Week</i> <i>(Dec 15 - 18)</i>	• <i>Thursday, December 18 at 12:00 PM</i>	